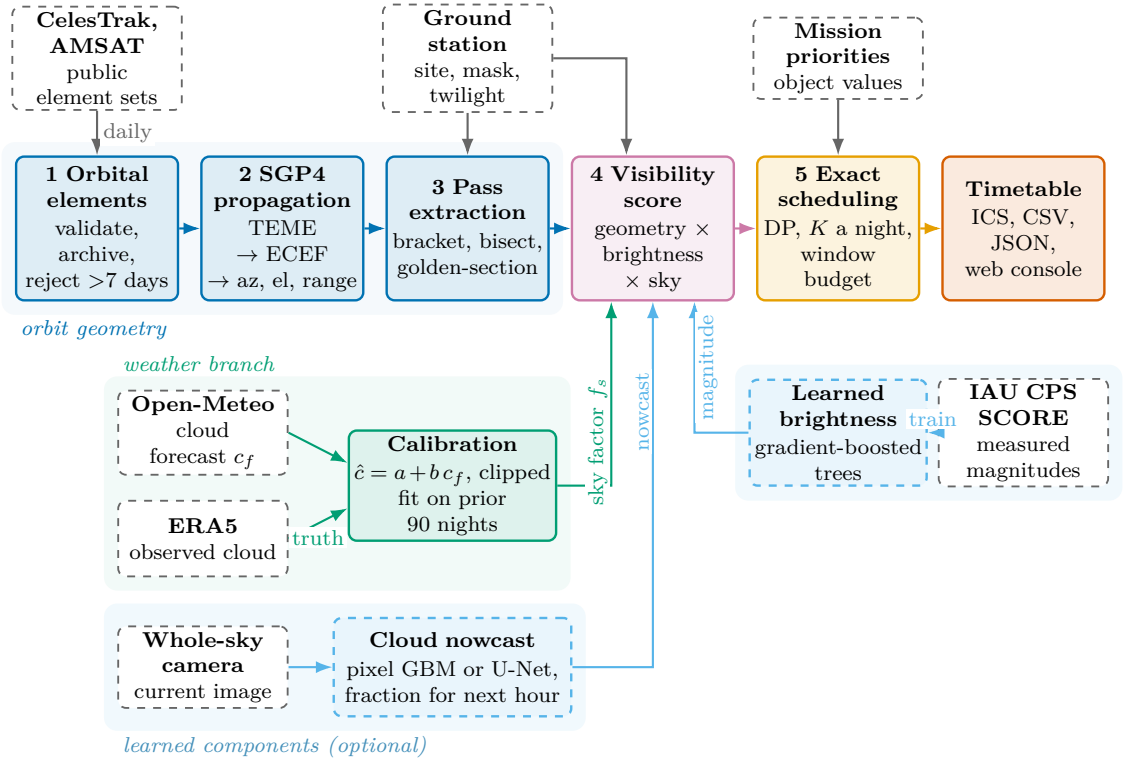
CelesTrak, AMSAT
public element sets
daily
Ground station
site, mask, twilight
Mission priorities
object values
1 Orbital elements
validate, archive, reject >7 days
2 SGP4 propagation
TEME → ECEF → az, el, range
3 Pass extraction
bracket, bisect, golden-section
4 Visibility score
geometry × brightness × sky
5 Exact scheduling
DP, K a night, window budget
Timetable
ICS, CSV, JSON, web console
orbit geometry
weather branch
Open-Meteo
cloud forecast c_f
ERA5
observed cloud
truth
Calibration
$\hat{c} = a + b\,c_f$, clipped
fit on prior 90 nights
sky factor f_s
nowcast
magnitude
Learned brightness
gradient-boosted trees
train
IAU CPS SCORE
measured magnitudes
Whole-sky camera
current image
Cloud nowcast
pixel GBM or U-Net, fraction for next hour
learned components (optional)